

CRT FIRST DAY

Q1. write a cpp program that declares a global int variable in s in size 200 in the main function takes an int input from the user. perform floating point division of the global variable by the user input using explicit type casting and store it in a local double variable nd print the result

```
sol: #include <iostream>

using namespace std;

int globalVar = 200;

int main()
{
    int userinput;
    cout<<"Enter an integer: "<<endl;
    cin>>userinput;
    double result = (double)globalVar/userinput;
    cout<<"the divsion is: "<<result<<endl;
    return 0;
}
```

Q2. write a cpp program that takes 3 integer input representing the side of triangles. print true if triangle is equilateral otherwise false using only relational and logical operators

```
Sol: #include <iostream>

using namespace std;

int main()
{
    int a,b,c;
    cout<<"Enter first integer: "<<endl;
    cin>>a;
    cout<<"Enter second integer: "<<endl;
    cin>>b;
```

```

cout<<"Enter third integer: "<<endl;
cin>>c;
if (a==b && b==c)
{
    cout<<"it is equivalent"<<endl;
}
else{
    cout<<"it is not equivalent"<<endl;
}
return 0;
}

```

Q3. Celsius to fahrenheit conversion

```

Sol: #include <iostream>
using namespace std;
int main()
{
    float f,c;
    cout<<"Enter celsius: "<<endl;
    cin>>c;
    f = (c * 9/5) + 32;
    cout<<"Temperature in fahrenheit is: "<< f <<endl;
    return 0;
}

```

Q4. write a cpp program that takes 2 integer input a and b multiply a *4 using bitwise left shift operator , print the value of a.

s=101

$10100 = x * 2^{\text{pow } n}$

left shift a= a<<2; (open)

right shift b=b>>3;(close)

```
Sol: #include <iostream>
using namespace std;
int main()
{
    int a,b;
    cin>>a>>b;
    a = a<<2;
    cout<<"value of a is: "<<a<<endl;
    return 0;
}
```

Q5. write a cpp program that takes 2 number & operator as input from user. perform the corresponding arithmetic operation and print relation

```
Sol: #include <iostream>
using namespace std;
int main()
{
    int a,b;
    char op;
    cin>>a>>b>>op;
    switch(op)
    {
        case '+': cout<< "addition : " << a + b;
        break;
        case '-': cout<< "subtract : " << a - b;
        break;
        case '*': cout<< "multiplication : " << a * b;
        break;
        case '/':
```

```

if (b != 0) {
    cout<< "division : " << a / b;
}
else {
    cout<<"cannot divide by zero";
}
break;
default: cout<<"invalid operator"<<endl;
}
return 0;
}

```

Q6. write a cpp program that takes 2 integer as input from the user, swap their values using a temporary variable.

Sol: #include <iostream>

```

using namespace std;
int main()
{
    int a,b;
    int temp;
    cout<<"enter your original integers: "<<endl;
    cin>>a>>b;
    temp = a;
    a = b;
    b = temp;
    cout<< "The values after swapping is : "<<endl;
    cout<<"a = "<<a<<endl;
    cout<< "b = "<<b<<endl;
    return 0;
}

```

Q7. write a cpp program that takes 3 int as input from user.

calculate their total sum and average and print them

Sol: #include <iostream>

using namespace std;

int main()

{

int a,b,c;

int sum;

double average;

cout<<"Enter your integers: "<<endl;

cin>>a>>b>>c;

sum= a + b + c;

average = (a+b+c)/3.0;

cout<< "the sum of the numbers: "<<sum<<endl;

cout<< "the average of the numbers: "<<average<<endl;

return 0;

}

Q8. password same then login else cannot login(2 values as input from user)

Sol: #include <iostream>

using namespace std;

int main()

{

int a,b;

cout<<"Enter your values: "<<endl;

cin>>a>>b;

if (a ==b)

{

cout<< "login successful"<<endl;

}

```

else
{
    cout<<"not correct"<<endl;
}
return 0;
}

```

Q9. q10. discount for student till 16 age group

17-64= no offer

```

Sol: #include <iostream>
using namespace std;
int main()
{
    int a;
    cout<<"Enter your value: "<<endl;
    cin>>a;
    if (a <=16)
    {
        cout<< "discount"<<endl;
    }
    else if( 17<=a <=64)
    {
        cout<<"cant"<<endl;
    }
    else
    {
        cout<<"not in that age group"<<endl;
    }
    return 0;
}

```

